PACKING $k^2 - \lceil\sqrt{32k}\,\rceil - 1$ UNIT SQUARES IN A SQUARE OF SIDE LESS THAN k

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ABSTRACT. Let $s(N)$ be the side of the smallest square containing N unit squares with disjoint interiors, and let $c^*(k)$ be the largest c with $s(k^2 - c) = k$. We prove $c^*(k) < 4\sqrt{2k} + 1$ for every integer $k \geq 6$, that is, $s(k^2 - \lceil\sqrt{32k}\,\rceil - 1) < k$. The construction is Ryu's L-shaped packing: an aligned grid and two bands of b tilted rows. Both ends of each band are filled with axis-parallel rows, and b is an odd integer within 1 of $\sqrt{2k}$. The odd width staggers the floors of the two lowest levels of each filled end, which is what the end estimate needs. The same idea and the same constant $4\sqrt{2}$ were found earlier by Evan Daniel, who proves $c^*(k) \leq 4\lceil\sqrt{2k-7}\,\rceil - 1$ for $k \geq 7$ in Lean; for $k \geq 89$ the two integer bounds differ by at most 2, and neither is the smaller at every k. This note is an independent second proof, with a different split of the free length and an odd band width. Exact rational certificates for 38 values of k up to 600 are checked by two separately written programs.

1. INTRODUCTION

For $x > 0$ let $N(x)$ be the largest number of unit squares that fit, with pairwise disjoint interiors and rotations allowed, in a square of side x. For an integer k, $N(k) = k^2$. Let

$$M(k) = \max\{N : s(N) < k\}, \qquad w(k) = k^2 - M(k), \qquad c^*(k) = \max\{c : s(k^2 - c) = k\}.$$

Since s is nondecreasing, $s(k^2 - c) < k$ exactly when $M(k) \geq k^2 - c$, so $c^*(k) = w(k) - 1$.

Erdős and Graham [EG75] showed that the waste $x^2 - N(x)$ is $O(x^{7/11})$, which makes $w(k)$ sublinear. The best exponent is now 3/5 [Bui25, McC26] (the earlier claim [CG20] contained an error [AB26]). For integer sides approached from below, Ryu [Ryu26] proved, among other things:

- $c^*(k) \leq 8\lceil\sqrt{k-4}\,\rceil - 1$ for every $k \geq 6$ (his Theorem 1.4), from an explicit L-shaped packing (his Theorem 1.3);
- $c^*(k) < 20.668\,k^{2/5} + 0.623$ for $k \geq 1.6 \cdot 10^7$ (Theorem 1.2), and bounds of order $k^{3/8}$ from $k \geq 2.48 \cdot 10^{12}$ (Theorem 1.1);
- computer-assisted bounds at five values of k between 10^5 and 10^8 (Theorem 1.6).

Daniel [Dan26b] proved in Lean, on 18 October 2026, $c^*(k) \leq 4\lceil\sqrt{2k-7}\,\rceil - 1$ for every $k \geq 7$, by filling the band ends of Ryu's L with axis-parallel layers, and $c^*(k) < 18.9\,k^{2/5} + 35.7\,k^{1/5} + 122$ for $k \geq 3 \cdot 10^5$ from Ryu's staircase ends. We learned of this after writing v0.1 of this note; Theorem 1 below is an independent proof of a bound of the same strength. Lower bounds of order $\log k$, $k^{1/4}$ and $k^{1/3}$ for large k are also due to Ryu [Ryu26b]. Arslanov, Mustafin and Shangitbayev [AMS21] proved $c^*(k) \leq k - 1$ for $k \geq 12$. In an earlier note [Fan26a] we proved $c^*(k) \leq k - 1 - F(k)$ for $k \geq 15$, with $F(k) = 1 + \max\{p + 2q : 16p + 21q \leq k - 15\}$.

Theorem 1. *For every integer $k \geq 6$, $w(k) < 4\sqrt{2k} + 2$. Equivalently $c^*(k) < 4\sqrt{2k} + 1$, so $c^*(k) \leq \lceil\sqrt{32k}\,\rceil$, and $s(n) < k$ for every integer $n \leq k^2 - \lceil\sqrt{32k}\,\rceil - 1$.*

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Draft v0.3, 2026-10-27, doi:10.5281/zenodo.23714459; all versions: doi:10.5281/zenodo.23624411. v0.3 corrects Sections 6 and 7 and some wording after the review [Lev26]; the mathematics is unchanged from v0.2 (doi:10.5281/zenodo.23624565), which credits the earlier result of Daniel [Dan26b]. Code and certificates: [Fan26b], folder `theorem-a/`, commit b1b3f6b.

The packing is Ryu's L (his Sections 3–4), and we use his end-region framework (his Section 6): lifted band ends, the far end as the 180° image of the near one, and end fills assembled with the rows. Right-justified axis-parallel layers in a band end already appear in Ryu's Section 7 staircase, after Bui [Bui25]. Daniel [Dan26b], and this note, fill each band end with such layers alone, at $b \approx \sqrt{2k}$. What differs here is the split of the free length (lift 0 at one end of each band, all of it at the other), the odd band width, and the estimate of Lemma 2, which together give the bound in the form $w(k) < 4\sqrt{2k} + 2$. To keep the note self-contained, we prove everything we use.

2. The band

Fix an integer $b \geq 3$ and a rational $t = \tan(\theta/2) > 1/b$. Write $c = \cos\theta = (1-t^2)/(1+t^2)$, $s = \sin\theta = 2t/(1+t^2)$, $T = s/c$, $\tau = c/s$, $\sec = 1/c$, and

$$W = bc + s \quad \text{(band width)}, \qquad h = bs + c \quad \text{(row height)}.$$

At $t_0 = 1/b$: $c_0 = (b^2-1)/(b^2+1)$, $s_0 = 2b/(b^2+1)$, $\tau_0 = (b^2-1)/(2b)$, $W_0 = b$, $h_0 = (3b^2-1)/(b^2+1) < 3$. Since $b - W = 2t(bt-1)/(1+t^2)$, we have $W < b$ for every $t > 1/b$.

Use band coordinates $X \in [0, W]$ (across) and $Y \in [0, L]$ (along). With (u, v) the coordinates along (c, s) and $(-s, c)$, the rectangle $P = \{0 \leq u \leq b,\ 0 \leq v \leq 1\}$ is the union of the b unit squares $\{i \leq u \leq i+1\}$. *Row* j is P translated so that its corner $u = v = 0$ is at $(s,\ y_0 + j\sec)$, for $0 \leq j < R$.

(R1) A point of P has $X - s = uc - vs \in [-s, bc]$, so every row lies in $0 \leq X \leq W$. *(R2)* The shift $(0, \sec)$ changes v by exactly 1, so in row 0's frame row j lies in $\{j \leq v \leq j+1\}$ and the rows have disjoint interiors. *(R3)* The lowest point of row j is its corner, at height $y_0 + j\sec$, and its highest is at height $y_0 + j\sec + h$. So the rows fit in $[0, L]$ when $y_0 + (R-1)\sec + h \leq L$. Let $R = R(L) = \lfloor (L-h)/\sec \rfloor + 1$, the most rows with $y_0 = 0$, and let $F = L - (R-1)\sec - h \in [0, \sec)$ be the free length.

End regions. With row 0 at lift y, let $\ell(X) = y + (X-s)T$ for $X \geq s$ and $\ell(X) = y + (s-X)/T$ for $X \leq s$. *(E1)* Every row lies in $\{Y \geq \ell(X)\}$: in row 0's frame a point of any row has $u, v \geq 0$, and $Y - y - (X-s)T = vc + vs^2/c \geq 0$ for $X \geq s$, while $Y - y - (s-X)/T = us + uc^2/s \geq 0$ for $X \leq s$. So a unit square in $\{0 \leq X \leq W,\ 0 \leq Y \leq \ell(X)\}$ meets no row in an interior point.

Fill. Let $a_i(y) = \max(0,\ s + (i+1-y)\tau)$ and $m_i(y) = \max(0, \lfloor W - a_i(y) \rfloor)$. Level i holds the m_i unit squares $[W - m_i + q,\ W - m_i + q + 1] \times [i, i+1]$, $0 \leq q < m_i$, and $\mathrm{fill}(y) = \sum_i m_i(y)$. *(E2)* These squares lie under ℓ. If $a_i \geq s$, ℓ increases on $[a_i, W]$ and $\ell(a_i) \geq i+1$. If $a_i < s$, then $i + 1 < y \leq \min \ell$.

(E3) The 180° rotation about $(W/2, L/2)$ maps the rows onto rows with lifts $y_1 + (R-1-j)\sec$, where $y_1 = L - y_0 - (R-1)\sec - h$. So the region over the last row is a rotated end region with lift y_1, and takes the rotated fill. The two fills are disjoint, since each vertical line $X = \text{const}$ meets row 0 and row $R-1$. A band of length L therefore holds

$$C(L) = bR + \mathrm{fill}(y_0) + \mathrm{fill}(y_1)$$

unit squares. Below, $y_0 = 0$ and $y_1 = F$.

3. The packing

Let $S = k - b + W < k$, and suppose $k - b > h$. Put three pieces in $[0, S]^2$:

- the aligned grid $[0, k-b]^2$;
- the band of length S in $[k-b, S] \times [0, S]$ (with $x = k - b + X$, $y = Y$);
- the band of length $k - b$, reflected in $y = x$, in $[0, k-b] \times [k-b, S]$.

They have disjoint interiors. Since $k^2 - (k-b)^2 = bk + b(k-b)$,

$$w(k) \leq b(k-S) + \big[bS - C(S)\big] + \big[b(k-b) - C(k-b)\big]. \tag{1}$$

4. The end lemma

From now on b is odd. Put $\eta = b/2 - \tau$; at t_0, $\eta_0 = 1/(2b)$. Say $t > 1/b$ is *close* if

(a) $\lfloor W - s - \tau \rfloor = (b-1)/2$;
(b) $0 \le \eta < 1/2$;
(c) $2\tau - bc \ge 0$;
(d) $h < 3$.

At t_0 we have $W_0 - s_0 - \tau_0 = (b-1)/2 + \delta_b$ with $\delta_b = \frac{1}{2} + \frac{1}{2b} - \frac{2b}{b^2+1} \in (0,1)$; here $\delta_b > 0$ because $b^3 - 3b^2 + b + 1 > 0$ for $b \ge 3$. Also $2\tau_0 - bc_0 = (b^2-1)/(b(b^2+1)) > 0$. So (a)–(d) hold strictly at t_0, and by continuity on an interval $(1/b, 1/b + \varepsilon_b)$.

Lemma 2. *If t is close, then* $\mathrm{fill}(0) \ge (b-1)/2$, *and for $0 \le y < \sec$,*

$$\mathrm{fill}(y) > \tfrac{b-1}{2} + 2z - 1 - 2\eta, \qquad z = W - s - 2\tau + y\tau.$$

Proof. $\mathrm{fill}(0) \ge m_0(0) = \lfloor W - s - \tau \rfloor = (b-1)/2$ by (a). For $0 \le y < \sec$ and $i \ge 0$, the clipping in a_i is inactive:

$$s + (i + 1 - y)\tau > s + (1 - \sec)\tau = c(1-c)/s \ge 0.$$

So level 0 holds at least $\lfloor z + \tau \rfloor$ squares and level 1 at least $\lfloor z \rfloor$. Since $\tau = \frac{b-1}{2} + \frac{1}{2} - \eta$ with $\frac{b-1}{2} \in \mathbb{Z}$, together they hold at least $\frac{b-1}{2} + \lfloor z + \frac{1}{2} - \eta \rfloor + \lfloor z \rfloor$. For $0 \le \eta < \frac{1}{2}$ and real z with fractional part f, the sum $\lfloor z \rfloor + \lfloor z + \frac{1}{2} - \eta \rfloor$ is $2z - 2f > 2z - 1 - 2\eta$ if $f < \frac{1}{2} + \eta$, and $2z - 2f + 1 > 2z - 1$ otherwise. □

The odd width matters: it puts the arguments of the two floors half a unit apart. For even b the arguments differ by an integer minus $1/(2b)$, and this pairing gives nothing.

Lemma 3. *If t is close and $L > h$, then with $R = R(L)$, $y_0 = 0$ and $y_1 = F$,*

$$bL - C(L) < bL(1-c) + bch - 2b + 2 - 2(W - s - 2\tau) + 2\eta. \tag{2}$$

At t_0 the right side equals $2bL/(b^2+1) + b + 2 - \kappa_b$, where $\kappa_b = (7b^4+1)/\big(b(b^2+1)^2\big)$.

Proof. $bR = b + bc(L - h - F)$. By Lemma 2, $C(L) > bR + (b-1) + 2(W - s - 2\tau) + 2F\tau - 1 - 2\eta$. The terms in F sum to $F(2\tau - bc) \ge 0$ by (c), so drop them and rearrange. The value at t_0 uses $1 - c_0 = 2/(b^2+1)$, $bc_0h_0 = 3b - b(10b^2+2)/(b^2+1)^2$, $W_0 - s_0 - 2\tau_0 = -(b^2-1)/(b(b^2+1))$ and $2\eta_0 = 1/b$. □

5. Proof of Theorem 1

Let $k \ge 6$ and let b be an odd integer with $|b - \sqrt{2k}| \le 1$. Then $b \ge 3$. For $k \ge 8$, $k - b \ge k - \sqrt{2k} - 1 \ge 3 > h$ by (d). For $k = 6, 7$ the only choice is $b = 3$, and $k - b \ge 3 > h$. For close t, (1) and (2) with $L = S$ and $L = k - b$ give $k^2 - n(t) < H(t)$, where $n(t)$ is the number of squares packed and $H(t)$ is $b(k - S)$ plus the two right sides of (2).

This holds for each t separately, even though R, F and the floors jump with t. H is continuous, and as $t \to 1/b^+$, $S \to k$ and

$$H(t) \to B(k) = \frac{4kb}{b^2+1} + 2b + 2 + \frac{2}{b^2+1} - 2\kappa_b.$$

Now $4kb/(b^2+1) < 4k/b$, and $4k/b + 2b = 4\sqrt{2k} + \frac{2}{b}(b - \sqrt{2k})^2 \le 4\sqrt{2k} + \frac{2}{b}$. So

$$B(k) < 4\sqrt{2k} + 2 - 2\Big(\kappa_b - \frac{1}{b} - \frac{1}{b^2+1}\Big),$$

and $b(b^2+1)^2\big(\kappa_b - \frac{1}{b} - \frac{1}{b^2+1}\big) = 6b^4 - b^3 - 2b^2 - b > 0$. Hence $B(k) < 4\sqrt{2k} + 2$. Some close t has $H(t) < 4\sqrt{2k} + 2$, and it gives a packing with $S < k$ and $k^2 - n(t) < 4\sqrt{2k} + 2$. □

Remark (Sharpness, observed). For odd $b \geq 5$ and $k = (b^2 - b + 8)/2$ the construction appears to have waste exactly $4b$, which is $4\sqrt{2k} + 2 - 31/(2b) + O(b^{-2})$; for example $k = 59$, $b = 11$, $w = 44$. We have no proof for every b. It holds at the t our program chooses for every odd b from 5 to 201, and the review [Lev26] found it in the limit $t \to 1/b^+$ for every odd b from 5 to 2001. Where it holds, the additive constant 2 cannot be lowered for this construction.

6. Checks and certificates

The construction is implemented in exact rational arithmetic (`lband.py` in [Fan26b]).

- *Admissibility.* The program takes $t = Q/(bQ - 1)$, so $t - 1/b = 1/(b(bQ - 1))$, for a target e: it starts at $Q = \lceil 2/(b^2 e) \rceil$ and halves Q until $k - S = b - W \geq e$. The target starts at $1/(10^6 k^3)$ and is divided by 10 until (a)–(d) and $H(t) < 4\sqrt{2k} + 2$ hold. It decides those conditions exactly, so every layout it returns satisfies the theorem's bound by Lemma 3.
- *A fixed target is not enough.* With $e = 10^{-7}$ at $k = 43218$ the program takes $b = 293$ and $t = 116/33987$, so $t - 1/b \approx 1.004 \cdot 10^{-7}$ and $k - S \approx 2.008 \cdot 10^{-7}$. That t is not close: (c) and (d) fail, while the close interval has width of order $b^{-3} \approx 4 \cdot 10^{-8}$. The waste there is $1178 = 4\sqrt{2k} + 2$, which misses the strict bound. Theorem 1 says nothing about such t.
- *Every step checked.* Every inequality in the proof was checked exactly for $6 \leq k \leq 20000$, and the bound for $6 \leq k \leq 60000$. The least slack $B(k) - w$ is $8019401/78023840599$, at $k = 19705$.
- *Certificates.* Strict certificates (every wall and pair margin positive) of the construction for $6 \leq k \leq 40$ and $k = 101, 300, 600$ (38 values) pass our checker and Evan Daniel's [Dan26]. The bundle ships 36 of them and rebuilds the two largest by script against published hashes. Both checkers were written with Claude. The largest has 359860 squares. A program that is not all-pairs runs both checkers unmodified on overlapping 4×4 windows: two unit squares whose centres are more than $\sqrt{2}$ apart cannot overlap, and both checkers accept such pairs without a separating-axis test.

The construction is not optimized. Searching over b, the number of rows and the split of the free length between the two ends gives waste at most $4\sqrt{2k} + 1.18$ for $15 \leq k \leq 300$; for example 9946 squares below side 100. A bounded search for better end and corner pieces by local optimization found gains of one or two squares only (at $k = 25, 30, 40$), not growing with b. That is consistent with the heuristic that a boundary between the aligned and the tilted lattice wastes about $1/2$ per unit length.

7. Comparison

The integer form of the theorem is $c^*(k) \leq \lceil \sqrt{32k} \rceil$.

- *Daniel's bound* [Dan26b], $4\lceil \sqrt{2k - 7} \rceil - 1$. For $7 \leq k < 10^6$ ours is smaller at 495210 values of k, his at 255215, and they tie at the remaining 249568. For $k \geq 89$ they never differ by more than 2; at 17 values of k below 89 his is smaller by 3 to 5. His $k^{2/5}$ bound is smaller from about $k = 7.4 \cdot 10^5$.
- *Ryu's Theorem 1.4.* Ours is smaller at $k = 6$ and for every $k \geq 9$; they tie at $k = 7$, and Ryu's is smaller at $k = 8$.
- *Ryu's Theorem 1.3*, optimized over b. Ours is smaller for $k \geq 14$.
- [Fan26a]. Ours is smaller from $k = 42$ (tie at 40, 41), and larger for $15 \leq k \leq 39$.
- *Superseded* by Ryu's Theorem 1.2 for $k \geq 1.6 \cdot 10^7$, by his Theorem 1.1 tiers beyond that, and by his computer-assisted values: at $k = 10^5, 10^6, 10^7$ they give $c^* \leq 1583, 3972, 10039$, against our $1789, 5657, 17889$.

So neither integer bound is the smaller at every k, and asymptotically Theorem 1 has the strength of Daniel's earlier bound. Against every bound in the Squares Project register, it is strictly the

smallest at 346919 values of k with $42 \le k \le 702000$ [Lev26]; we get the same count against Daniel's two bounds, Ryu's Theorem 1.4 and his value at $k = 10^5$.

Acknowledgements and credit

The L-shaped packing, its end-region framework and most of the bounds compared above are Sungjoon Ryu's. The axis-layer filling of the band ends and the bound $4\lceil\sqrt{2k-7}\,\rceil - 1$ were found first by Evan Daniel [Dan26b], whose certificate format and checker we also use. The Squares Project (Joshua Levy) keeps the register in which these bounds are compared. The construction code, checkers and certificates were written and run with Claude (Anthropic) as a coding and research agent, directed and reviewed by the author. Two blind language-model reviews of the proof (Claude Opus; OpenAI Codex, GPT-6 Astra) found errors that were corrected. The Squares Project's review [Lev26], a re-derivation by Claude from the note alone, accepted the proof and found the errors in Sections 6 and 7 that v0.3 corrects. None of these is a referee's report.

References

[AB26] M. Z. Arslanov, H. D. Bui, Note on "Efficient packings of unit squares in a large square", Discrete Comput. Geom. 76 (2026), 805–812, doi:10.1007/s00454-025-00767-w.

[AMS21] M. Z. Arslanov, S. A. Mustafin, Z. K. Shangitbayev, Improved packings of $n(n-1)$ unit squares in a square, Electron. J. Combin. 28(4) (2021), P4.22, doi:10.37236/8586.

[Bui25] H. D. Bui, Square packing with $O(x^{0.6})$ wasted area, arXiv:2508.04603 (preprint; v2, 15 March 2026).

[CG20] F. Chung, R. Graham, Efficient packings of unit squares in a large square, Discrete Comput. Geom. 64 (2020), 690–699, doi:10.1007/s00454-019-00088-9.

[Dan26] E. Daniel, `https://github.com/evand/square-packing`, `search/exact/verify_cert.py` at commit `29d0cee87c1d717f7b4497b8fb0ee17ac9d0087d`.

[Dan26b] E. Daniel, Upper bounds on $c^*(k)$, in Lean, `https://github.com/evand/square-packing`, `certificates/k2-minus-c/` at commit `aa12b98` (18 October 2026); write-up `https://evand.github.io/square-packing/k2c/`.

[EG75] P. Erdős, R. L. Graham, On packing squares with equal squares, J. Combin. Theory Ser. A 19 (1975), 119–123, doi:10.1016/0097-3165(75)90099-0.

[Fan26a] K. Fang, Packing $n^2 - n + 1$ unit squares in a square of side less than n, for every $n \ge 15$, draft v0.9 (2026), doi:10.5281/zenodo.23463481.

[Fan26b] K. Fang, squarepack-certs, `https://github.com/TheSnakeFang/squarepack-certs`, folder `theorem-a/`, commit b1b3f6bda13739c9eeecc1ceaefec9178294e0ca.

[Lev26] The Squares Project (J. Levy), review of v0.2 of this note, 25 October 2026, `https://github.com/jlevy/squares/blob/main/docs/project/reviews/review-2026-10-25-fang-theorem-a.md`; issue `https://github.com/jlevy/squares/issues/797`.

[McC26] R. McClenagan, Optimally packing a large square by unit squares, arXiv:2602.01484 (preprint, 2026).

[Ryu26] S. Ryu, Packing $k^2 - c$ unit squares: an upper bound of order $k^{3/8}$ for the deficiency, preprint, version 1.1 (10 October 2026), doi:10.5281/zenodo.23278927 (version 1.0: doi:10.5281/zenodo.23256655).

[Ryu26b] S. Ryu, Packing $k^2 - c$ unit squares: $s(k^2 - c) = k$ for all large k, preprint, version 1.2 (2026), doi:10.5281/zenodo.23194104; and the companion preprints on lower bounds of order $k^{1/4}$ (doi:10.5281/zenodo.23212643) and $k^{1/3}$ (doi:10.5281/zenodo.23213564).

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